CO1 - AT1 - SHORT ANSWERS

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Course Code: ITA0502

Course Name: Computer Vision

Sampling: Low spatial resolution means the image has too few pixels. small objects lose details.

Improvement: Increase the sampling rate / use a higher resolution camera to capture more pixels

Night Image enhancement: use histogram equali-zation and noise Filtering Histogram equalization improves brightness and contrast, while Filtering reduces noise

Quantization: Quantization converts continuous intensity values into a fixed no of digital levels. It makes the image suitable for digital storage and processing.

Uneven lighting: use histogram equalization or adaptive contrast enhancement. It improves contrast in dark and bright areas, making product defects easier to see

5. Image segmentation : Divide the image into regions such as road, vehicles and pedestrians. This helps the car understand its surroundings and make safe driving decisions

6. Edge detection: Apply an edge detector such as canny edge detection. Cracks create strong Intensity changes. so their edges can be detected and identified

7. Filtering: Apply a median Filter to remove unwanted noise while preserving important tumor boundaries. This makes the tumor easier to identify

8. object detection: use an object detection method such as YOLO to identify vehicles and track their positions. It helps in traffic counting, congestion detection and traffic management

Blurred image : The possible cause is motion blur due to movement during image capture use deblurring / restoration techniques or a faster shutter speed to improve the image.

10. Image classification: Train a classifier to categorize products as defective or good based on their visual features. This supports automatic quality inspection and reduces manual checking.